Ilia Zlobin

Forward Deployed Engineer at Google

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Forward Deployed Engineer at Google with 14+ years of experience designing and building large-scale distributed systems, infrastructure platforms, and developer tooling. I specialize in turning ambiguous, high-impact problems into scalable technical solutions, setting architecture and technical direction while staying deeply hands-on through prototyping, implementation, and production rollout. My experience spans platform engineering, cloud infrastructure, developer productivity, and AI/ML systems, with a strong track record of leading multi-team initiatives, simplifying complex environments, and delivering measurable improvements in reliability, engineering velocity, and operational efficiency. Increasingly focused on applied AI and agentic systems, where I combine systems engineering, rapid technical discovery, and cross-functional collaboration to move ideas from early prototypes to production-scale solutions.

Google (Forward Deployed Engineer) **[October 2026 - Present]**

Meta, Bay Area, CA (Software Engineer) **[March 2026 - September 2026]**

- Architected and built a large-scale infrastructure fulfillment simulation engine (Python/C++), running ~50,000 simulations per week to evaluate capacity-plan feasibility and robustness under supply and demand uncertainty.
- Designed a generalized scenario-modeling platform for capacity fulfillment, covering 3+ classes of planning scenarios and integrating with internal orchestration and UI workflows for end-to-end what-if analysis at production scale.
- Designed and built a Rust-based meta-harness unifying 3 coding-agent runtimes behind a common orchestration and control layer, enabling centralized management of concurrent AI-assisted engineering workflows.

Mastercard, New York, NY (Principal Software Engineer) **[October 2024 - March 2026]**

- Architected an organization-wide access and permission management framework across 15 internal platforms and 10,000 users, establishing least-privilege IAM, policy-as-code guardrails, access lifecycle management through GitOps.
- Led the re-architecture of the internal developer platform from a custom Java solution to Backstage, defining the target architecture and migration across 10 teams and 20+ systems while enabling governance of 20,000+ technical assets.
- Led hands-on redesign of company-wide developer onboarding across multiple platform teams, reducing onboarding time by ~80% and driving a 2× increase in portal registrations the following quarter.
- Designed a self-service ephemeral development environment platform, reducing prototyping time by 80% and accelerating feature delivery by 30%.

EPAM Systems, Jersey City, NJ (Systems Architect) **[August 2021 - September 2024]**

- Designed and built a multi-tenant AWS Landing Zone infra management framework with Terraform/Spacelift, reducing setup time by 70% and ensuring SOC 2 compliance, across 250+ accounts.
- Delivered a MySQL cloning solution (Go/NextJS) providing developers with a self-service portal for on-demand database clones handling over 100 daily clones, accelerating application feature development by 80%.
- Authored infra & Kubernetes design for a GKE modernization factory, migrating 200 services to a 2,000-node platform.
- Spearheaded a cost optimization initiative for GKE workloads through data pipeline (Go, Cloud Functions), automated data analysis, and establishing rightsizing policies with guardrails, all resulting in over $1M in annual cost savings.
- Drove the adoption of serverless patterns (KNative, CloudFunctions, CloudRun) used by more than 10 teams.
- Led active-active DR architecture with DynamoDB global tables & Step Functions, ensuring near-zero downtime failover.
- Authored infrastructure and Kubernetes aspects of the GKE modernization factory, resulting in the migration of 200 services over 6 months from GCE to a global GKE platform, operating at a scale of 2,000 nodes over 3 regions.
- Migrated 100 microservices (Python, Go) from GCP to AWS on a new EKS platform using Sceptre, CloudFormation, and FluxCD with Linkerd, adding OpenTelemetry and MySQL replication for zero-downtime cutover.
- Designed and implemented a secure identity federation pattern (Python) enabling critical financial applications in AWS to securely consume GCP services, extending the model for the release pipelines through GitHub Actions.

EPAM Systems, Minsk, Belarus (Lead DevOps Engineer) **[October 2018 - August 2021]**

- Built a multi-region Kubernetes platform (200+ microservices, Istio, GitOps), reducing deployment time 80% and ensuring 99.99% availability.

- Implemented a multi-cloud landing zone using Cloud Posse's Atmos, enabling consistent provisioning across AWS, Azure, and GCP. Achieved 70% faster bootstrapping and unified management for 300+ accounts, enhancing operational efficiency and security.
- Implemented a GitOps workflow with FluxCD and progressive deployment with Prometheus metrics. This approach decreased deployment errors by 80% and improved lead time by 70%.
- Engineered a high-performance Terraform drift detection system in Go, processing configurations across 200+ repositories and 1000+ resources daily.
- Built an EKS-based platform with advanced observability (Prometheus/Grafana) and custom blue-green deployment support, reducing mean time to recovery (MTTR) by 60% and improving overall system reliability by 40%.
- Designed and implemented a unified observability system using the ELK stack and Prometheus/Grafana with APM capabilities, enabling DevOps KPIs and reducing Mean Time to Detect (MTTD) issues by 50%.
- Developed a custom Kubernetes operator (Go) to automate database operations, reducing manual DBA interventions by 80% and improving database reliability.

NIIAS, Saint Petersburg, Russia (Software Developer) **[October 2012 - August 2018]**

- Developed and maintained mission-critical railway control embedded software using C++ with real-time Linux OS, ensuring 99.99% uptime for systems managing 1000+ daily train operations. Optimized core algorithms, resulting in a 40% reduction in CPU usage and 50% improvement in response times.
- Led the migration from monolithic architecture to containerized microservices using Docker, enhancing system modularity and portability. This initiative improved deployment frequency by 300%, reduced time-to-market for new features by 50%.
- Implemented automated testing practices, increasing code coverage from 60% to 95% and reducing post-release defects by 80%. Introduced behavior-driven development (BDD) practices, improving collaboration between developers and business stakeholders.
- Engineered an embedded video streaming system in C++ for 100 trains with real-time Linux OS, optimizing for constrained hardware to deliver 720p@30fps with <1s latency (99th percentile) over 3G networks.
- Developed a centralized train control center software using C++ and Qt, enabling remote operation of multiple trains with telemetry visualization, predictive collision alarms, and digital location map.

Education

Bachelor's Degree in Computer Science (with honors) St. Petersburg State Transport University, 2012

Professional Certifications:

- **Professional** Cloud Architect, **Google Cloud**
- **AWS** Certified Solutions Architect – **Professional**
- **AWS** Certified DevOps Engineer - **Professional**
- **Microsoft** Certified: Azure Solutions **Architect Expert**
- **CKA**: Certified **Kubernetes** Administrator

Top Projects **[December 2018 - Present]**

- **Hermes: Autonomous AI Agent Fleet –** Built and operated an autonomous multi-agent fleet on the open-source Hermes Agent framework with 98 specialized worker profiles across 21 dependency-gated workflows, completing 2,100+ tasks at a 93% completion rate. Case Study
- **AI Events concierge (Full-Stack Agentic System)** – Designed and implemented an AI concierge for event discovery and registration using multi-agent workflows. Automated search, ranking, form submission, and calendar integration, enabling end-to-end scheduling with minimal human input. GitHub (Video)
- **Event ingestion and ranking pipeline (Full-Stack Project**) – Built a serverless data processing platform (AWS/SST) for ingesting and ranking social events with crawlers. Implemented hybrid search and ranking with OpenSearch and automated publishing to external platforms, supporting analytics and real-time discovery at scale. GitHub
- **DSPy: Declarative Language Programs (AI/ML Research)** – Explored **DSPy**, a framework for declarative LLM pipelines. Built notebooks and demos for workflow orchestration, prompt optimization, and evaluation. GitHub (Video)
- **Speech analysis and visualization with ML (Full-Stack Project)** – Developed toolkits and full-stack applications for pronunciation and pitch analysis, deployed on AWS SageMaker and serverless backends. Delivered real-time visual feedback for learners via modern web frontends. GiHtub
- **LLM Fine-Tuning (ML Research)** – Explored optimization strategies for customizing transformer models, covering quantization, PEFT, and evaluation frameworks. Delivered reproducible research workflows and benchmarks on GPU infrastructure. GitHub (Video)
- **Atmos Landing Zones (IaC Project)** – Built secure, scalable AWS Landing Zones with Cloud Posse Atmos, Terraform, and Helmfile. Automated multi-account provisioning, IAM delegation, centralized networking, security guardrails, audit logging, and Kubernetes orchestration. Enabled CI/CD, reproducible developer environments, and extensible configs for future cloud expansion. GitHub